

Understanding that compounds through work

Phossil proposes an operating and learning layer that connects the information, relationships, and reasoning behind work across people, teams, systems, and time. Its purpose is not only to preserve understanding, but to help people use it to explore, develop ideas, coordinate, decide, act, and learn from what actually happens.

Research-backed position paper and evaluation agenda. Phossil, September 2026. Version 0.1 for review. The product mechanisms and cases below are proposals, not claims of generally available capability or demonstrated customer results.

1. The argument

Work produces more than deliverables. People develop an understanding of customers, constraints, relationships, alternatives, and consequences. They discover which questions matter, which assumptions no longer hold, and where an earlier approach may be useful again. That understanding can shape several initiatives and survive longer than any individual task.

Yet a collection of messages, documents, tickets, and completed tasks is not equivalent to a usable account of that experience. A decision may be recorded without its conditions. An issue can be closed without establishing the intended outcome. A person may appear frequently in a discussion without having authority to approve its conclusion. A plausible explanation can become accepted through repetition rather than testing.

Phossil's thesis is that people and organizations could benefit from an evolving account of work that preserves these distinctions and remains connected to the tools where work happens. This account would not be a definitive model of the organization. It would be a permitted, inspectable, time-sensitive interpretation: useful enough to act with, explicit enough to question, and revisable when circumstances change.

The ambition spans individual work, human collaboration, and work involving AI. An individual might use the same underlying understanding to investigate a question, develop an idea, or reconsider a commitment. A team could examine dependencies and make a decision. A leader could compare local explanations without prematurely treating them as an organization-wide diagnosis. An AI assistant could use relevant context while staying within a delegated task. These are related uses of a proposed capability, not separate promises that have already been validated.

The central hypothesis is testable: maintaining the relationships between sources, interpretations, decisions, actions, and outcomes can improve the quality and continuity of work enough to justify the effort of maintaining them. The relevant comparison is not an empty folder or an unaided worker. It is

a well-configured combination of existing work tools, search, AI assistance, and normal review practices.

Three conditions follow. First, people must be able to inspect and correct what Phossil says. Second, understanding must help work progress in existing tools, not become another reporting obligation. Third, the system must distinguish a proposed next step from an authorized action, and a completed action from a verified result. If these conditions are not met, a richer organizational model could merely create more convincing mistakes.

2. What the research supports - and does not

The following evidence was selected to test the rationale, not to calculate a projected Phossil return on investment. It includes favorable findings, counterevidence, and research on collaboration and learning. The studies differ in populations, interventions, and outcomes; their percentages must not be pooled into a single productivity claim.

Assistance can help, unevenly

Brynjolfsson, Li, and Raymond's November 2024 revision studies a staggered AI-assistant deployment involving 5,172 customer-support agents. Issues resolved per hour increased by 15% on average. Benefits varied with experience; the highest-skilled and most-experienced workers saw small quality declines alongside limited speed gains. This is a deployment study in one setting, not an individually randomized trial or evidence for every kind of knowledge work. [1]

For Phossil, the design implication is a hypothesis: assistance should adapt to context and expertise, and quality must be measured alongside throughput. The study does not establish that persistent organizational understanding caused the gains, or that Phossil would reproduce them.

Perceived speed is not measured performance

METR's early-2025 randomized study covered 246 tasks performed by 16 experienced open-source developers in familiar repositories. Allowing AI increased completion time by 19%. The sample, tools, and tasks were specific; this is not a current universal estimate of AI's effect. [2] In February 2026, METR said newer tools likely offered greater speedup, but selection effects made its follow-up data unreliable for estimating the magnitude. [3]

This counterexample matters because an assistant can feel useful while increasing total review and correction work. Phossil should measure whole-task outcomes, including verification and rework, rather than use engagement, generation volume, or confidence as a substitute for value.

Collaboration structure matters

Yang and colleagues examined collaboration during Microsoft's shift to firm-wide remote work. Their analysis found more siloed collaboration and less connection across groups. The study concerns a particular company and pandemic-era transition; it does not establish that remote work generally reduces productivity. [4]

Our inference is narrower: the connections through which people encounter information deserve attention. An interface that only summarizes the current user's documents may miss how another group's work changes the meaning of those documents. Whether Phossil can surface such connections accurately and without excessive interruption remains an empirical question.

Experience does not automatically become transferable learning

Darr, Argote, and Epple analyzed 18 months of weekly data from 36 pizza stores owned by 10 franchisees. The publisher abstract reports learning, knowledge depreciation, and transfer among commonly owned stores but not across different owners. This observational, historical service setting cannot predict modern software outcomes. It does show why accumulation and transfer should be treated as conditional rather than automatic. Only the abstract and bibliographic record were inspected for this paper. [5]

For Phossil, retaining an episode is therefore not enough. Reuse must preserve the conditions under which an interpretation was useful and give the next person a way to decide whether those conditions still apply.

Governance is an ongoing practice

NIST's AI Risk Management Framework 1.0 organizes voluntary risk management around Govern, Map, Measure, and Manage. We use the January 2023 version as a reference for lifecycle-oriented risk thinking, not as a certification or a claim of compliance. [6]

Together, these sources justify investigation, not a conclusion that another software layer is necessary. They leave unresolved the cost of maintaining an account, the accuracy of cross-system interpretation, willingness to correct it, and the incremental benefit over tools already available.

3. A proposed operating mechanism

The unit of value is not a summary. It is a useful relationship between what is known, what someone is trying to do, what they choose, and what happens next. Phossil proposes to maintain those relationships across changing work without claiming to know more than permitted sources establish.

Keep different kinds of knowledge distinct

An observation records something attributable: a statement in a message, a document revision, or a test result. An interpretation explains what it may mean. A question identifies something unresolved. A decision records a choice, its scope, and the applicable authority. An action records what was attempted or completed. An outcome records an observed consequence over an identified period. Reviewed learning records what participants judge worth carrying forward, together with limits and counterexamples.

These distinctions are consequential. A message saying a launch is ready is an attributable statement, not independent proof of readiness. A passed test establishes only the conditions tested. Two initiatives sharing a specialist may suggest a capacity dependency but does not prove that this

dependency caused delay. Disagreement should remain visible until resolved, superseded, or explicitly retained.

Keep work in its working tools

The experience can begin in a messaging conversation, a document, an issue, or a user-invoked Phossil surface. A relevant message might receive a concise explanation with source links and an option to inspect further. People should be able to reply, clarify, or correct in that conversation. They should not have to open a new application for every question.

A Flow Card is a user-invoked desktop panel that brings relevant context and next-step choices alongside the current work. It is not the Workspace compressed into a card. Explorer combines an AI conversation with an interactive, fixed-size canvas for examining questions, relationships, alternatives, and sources. Workspace provides deeper shared work context, including evidence, options, decisions, and their history. A person can enter deeply when needed and return to the source tool without losing the thread.

Not every episode must visit all three surfaces. A simple clarification may finish in a message. A complex disagreement may require Explorer and a shared Workspace. The surfaces must share state and meaning, rather than maintain independent copies of the same account.

Make sources usable, not decorative

Every source reference should identify the originating application when known, the actual record type or file format, its title, and the relevant date or version. A provider logo is a recognition aid, not evidence of reliability. A citation should open the precise message, passage, page, or file where possible; otherwise it should open the record and explain the locator limitation. Unknown origin or format should stay unknown rather than receive an invented logo or extension.

Interpretations must show which source supports or challenges them. Large collections need searchable inspection, not a wall of badges. A conflicting source should not disappear behind a count. If access changes, the system must re-evaluate what each viewer may see, including derived summaries and citations, rather than preserve a restricted excerpt in a more accessible surface.

Move from understanding to action without inventing authority

Phossil may propose options, prepare a draft, or help someone locate an approver. It must not infer decision rights from activity or seniority alone. Approval should reuse a current, applicable authority basis when available. Missing authority should pause the affected action and give a direct next step, not interrupt unrelated exploration.

Authorized changes should occur through the relevant working tool and return attributable completion or failure information. A request to update an issue is not proof that it was updated. A completed change is not proof that the intended result followed. This separation is essential to any later learning.

4. The alternatives already do substantial work

Phossil does not start from a market without context, memory, or workflow. Existing products address overlapping parts of the proposal. The following comparison uses current first-party descriptions, inspected in September 2026. These establish documented positioning and features, not independently measured effectiveness or exhaustive product limitations.

Productivity, project management, and issue tracking

Task and issue systems provide durable assignments, status, dependencies, and execution history. Asana describes AI Teammates using its Work Graph, shared memory, and governed workflows. [7] Atlassian documents Teamwork Graph connectors supplying cross-tool context to Rovo Search, Chat, and Agents. [8] It would be inaccurate to describe these categories as mere to-do lists or to claim they lack connected context.

The proposed Phossil distinction must therefore be demonstrated at the level of an episode: can a correction to an interpretation change the right advice across tools and later work, while preserving scope, authority, and history? A feature label such as graph, memory, or agent does not settle that comparison in either direction.

Enterprise search, knowledge management, and AI context

Glean describes an Enterprise Graph connecting people, teams, projects, products, and processes, including task-oriented context for agents. [9] Microsoft documents permission controls for content brought through Copilot connectors. [10] Workfabric describes cross-application sensing, operating context, and living digital twins for agent use. [11] These are meaningful overlaps, not peripheral competitors.

Knowledge repositories and search remain valuable for storing and finding material. Phossil should not recreate their basic strengths. Its hypothesis concerns whether an evolving, correctable account adds value when a person must interpret what was found, develop or compare options, coordinate a response, and later examine the result. That is a proposed distinction, not an established exclusive capability.

Messaging, documents, and workflow automation

Messaging carries questions and informal commitments; documents support expression and revision; workflow systems route events and actions; business systems retain operational records. These are both potential complements and substitutes. An organization may solve a particular problem through better templates, clearer ownership, existing integrations, or disciplined reviews without introducing Phossil.

A fair evaluation must include that possibility. If a well-configured existing stack achieves equivalent quality and continuity with less effort, an additional layer is not justified for that use. Phossil would need to earn adoption through demonstrated incremental value, not the breadth of its vocabulary.

5. Case one: developing a new partner service

This is a fictional proposed workflow, not a Visa account or a customer result. A business-services company is exploring an onboarding service for partners. The case begins with an opportunity, not a known defect: a partner-marketing lead asks in a shared messaging channel whether customers would benefit more from a reusable starter kit or guided setup sessions.

The initial situation

Marketing understands recurring customer questions. Product knows what the service can reliably support. Partner managers know differences between accounts. Operations knows the effort required to deliver guided sessions. A commercial sponsor controls the experiment budget; none of these roles automatically owns the others' decisions.

Permitted inputs include a research-notes document, a draft service description, a support-topic report, an issue tracking invitation setup, and a planning spreadsheet. The report aggregates support categories but does not establish why people struggled. The spreadsheet estimates staffing effort rather than measuring completed delivery. The document emphasizes self-service while partner notes describe requests for help. These are different kinds of information, not interchangeable proof of demand.

From a message to an investigation

A proposed Phossil reply identifies the two approaches and links the relevant records. It asks whether the team wants to compare them against the same intended customer outcome. The lead can clarify in the channel: the aim is not to maximize guide downloads but to help a new partner complete its first useful setup.

Opening a Flow Card alongside the service document reveals that clarified aim, the current options, and the unanswered question about delivery effort. The lead can return to writing or open Explorer for a deeper comparison. Explorer retains conversation beside the canvas; it does not replace the investigation with a different, disconnected page.

The canvas separates customer needs, candidate service approaches, capability constraints, and assumptions. The lead can ask whether a hybrid approach makes sense or introduce another idea. Source links remain attached to specific statements. A correction that one interview concerned an experienced administrator changes the relevant interpretation; it does not silently rewrite all partner needs or alter the original interview.

A bounded decision, followed by work

The team uses Workspace to compare a starter kit, guided sessions, and a hybrid trial. The sponsor approves a limited trial; Product reviews capability wording; Operations confirms available delivery effort. Phossil records the scopes separately. It does not convert one approval into general permission to launch, spend, or make new commercial claims.

The lead revises the service document in the document editor. Operations creates session slots in its scheduling tool. Product updates the invitation issue through its normal process. Phossil may prepare or execute only the changes that have been authorized and supported by enabled integrations. It links their actual records back to the decision, including failures or work still pending.

What the team can learn

The trial plan specifies what will be observed: completion of the first setup, assistance required, operating effort, and reasons for abandonment. Downloads and attendance alone cannot establish successful onboarding. An improvement among experienced administrators would not demonstrate success for first-time users or all partner types.

After the trial, participants review what happened and what remains unexplained. They may retain a useful distinction between setup roles, reject an assumption about self-service, or decide the question is still open. A later service initiative can inspect that reasoning and test whether its audience and conditions are comparable. Reuse is a considered application, not an automatic rule.

How this case could fail

Phossil could mistake vocal requests for broad demand, favor older material with more documentation, or collapse several roles into a single owner. It could also create more review work than the team saves. Evaluation should compare the quality of the experiment decision, correction accuracy, total preparation effort, and the validity of later reuse against the team's configured existing tools. No benefit is claimed here.

6. Case two: an AI-assisted change after requirements shift

This fictional case concerns a software team adding bulk invitation capabilities. Work begins in an issue tracker. An AI coding assistant is helping an engineer implement the change when a product conversation clarifies that administrators must be able to cancel pending invitations. The original issue and tests do not cover cancellation.

The initial situation

The engineer has access to the repository, issue, product specification, and relevant shared messages. A support thread explains one customer's concern but is not permission to generalize the requirement to every account. Product can approve intended behavior. Repository rules determine code review and merge authority. Release approval remains a separate responsibility.

The evidence is not simply missing information. It is a temporal mismatch: one source describes an earlier requirement, another proposes a change, and implementation is already in progress. The coding assistant may produce internally consistent code that no longer answers the team's current question.

Make the mismatch inspectable

In the product conversation, Phossil could link the newer clarification to the affected issue and ask whether the interpretation is correct. If the engineer invokes a Flow Card beside the editor, it presents the old condition, the new statement, and what has not yet been approved. It does not declare every ongoing implementation obsolete.

The engineer can correct the scope: cancellation is required for invitations that have not been accepted, not for active users. This correction must change subsequent advice, affected tests, and any proposed work package. A summary that repeats the old scope after correction would be a material failure, even if its citations remain clickable.

Explorer supports deeper reasoning when alternatives matter. The engineer and Product can compare cancellation before delivery, cancellation before acceptance, and an expiry-based alternative. The canvas shows conditions and consequences alongside the chat. Workspace retains the shared options and decision; unrelated issues and private notes remain outside the account.

Keep execution in the engineering workflow

Once Product approves the intended behavior, the issue and specification can be updated through authorized actions. The AI assistant receives a task-scoped context package with the applicable requirement version, repository constraints, relevant decision, and unresolved questions. Content from a source document is data, not permission for the assistant to change its instructions or use unrelated tools.

The engineer works in the editor, commits on the appropriate branch, and opens a pull request. Normal automated checks and independent review still apply. Phossil should expose mismatches or stale assumptions, not become an alternative route around repository controls. If tests fail, it reports the failure; if the merge is approved, it records that event without calling the feature released.

Separate completion from results

After an authorized release, operational observations may show whether cancellation works under the intended conditions. Passing unit tests is useful but does not establish that customers understand the interface. A rollback, support confusion, or an untested edge case can change the account. The history should preserve the earlier decision and the later evidence rather than replacing one with the other.

Reviewed learning might concern a requirement pattern, a test gap, or a recurring handoff problem. Work Memory would retain how the episode unfolded. Decision Memory would retain the choice, alternatives, conditions, and authority. Neither should convert a local implementation choice into a company-wide engineering rule without review.

How this case could fail

The system could apply the wrong requirement version, omit a changed condition from the AI handoff, or treat a product approval as release authority. Source material might contain malicious instructions. Even correct context could increase engineering time. Tests should therefore include adversarial source content, stale and revoked access, deliberate corrections, failed actions, and reopening the work later. Measure total completion and review time, accepted quality, and incorrect recommendations after correction, not generated lines of code.

7. Case three: investigating a portfolio without inventing a single cause

This fictional case begins when a marketing VP asks in a leadership channel why three initiatives appear healthy in status reports while major commitments keep moving. A director oversees partner programs; individual leads manage a launch, an enablement effort, and an event. The question is ambiguous. No common cause has been established.

Preserve different scopes

The VP's view concerns commitments across the portfolio. The director's concerns include sequencing and shared resources. Each lead knows local delivery conditions. A single hierarchy is insufficient: someone may contribute to one initiative, decide within another, and review a third. Phossil must retain responsibility in context rather than infer that the most senior participant decides everything.

Permitted inputs include planning records, status messages, project issues, and an approved capacity plan. The plan describes expected allocation, not actual time worked. A private performance note is excluded. A missing allocation record remains a gap; it cannot be reconstructed by counting messages or monitoring individual activity.

Explore separate issues together

Phossil could respond in the channel with the observed commitment changes and the limits of the available account. A Flow Card offers a concise overview and a deeper exploration. Explorer opens the three initiatives side by side, preserving each original conversation and question. Looking for a relationship does not merge them into one issue.

The canvas shows possible connections: two initiatives may depend on the same specialist; one may have changed its definition of ready; another may be affected by a vendor date. Users can zoom and pan without resizing the surrounding canvas window, inspect a connection, or return to one initiative. Source records are available through selection rather than an unreadable network of every possible edge.

The director corrects one assumption: the shared specialist is only advising the event, not doing its delivery work. That correction removes the unsupported workload inference. It does not erase the person's involvement, change other assignments, or settle whether the remaining two initiatives compete for capacity. The system should make the affected interpretations visible.

Decide what can actually be decided

Workspace can hold a comparison of sequencing options, named commitments, and unresolved dependencies. The director might be authorized to propose a sequence but need initiative sponsors to accept changed dates. A missing approval should be a direct, local next step. Unrelated investigation should continue.

The resulting actions remain in planning tools, issue trackers, and team channels. Owners update their commitments; affected colleagues receive scoped explanations and links. A private explanation must not be copied into a broader announcement. A failed update leaves the action pending rather than falsely synchronizing the portfolio.

Review consequences at the right scale

The team later checks whether the chosen sequence reduced contention, whether commitments became more reliable, and whether work quality or customer outcomes worsened elsewhere. A date being met does not prove the original diagnosis. Other conditions may have changed; estimates may have been generous; work may simply have been deferred.

The reusable learning could be a planning question worth asking when initiatives share a specialist, not a rule that shared specialists always cause delay. Another department should be able to inspect the episode's conditions and decide whether the reasoning applies. Broader organizational understanding grows through such qualified connections, not by averaging status colors.

How this case could fail

Phossil could manufacture a common cause, expose restricted personnel context, or reward teams with more documentation. It could encourage leaders to mistake an interpretive map for complete organizational visibility. Evaluation must test whether it preserves genuinely unrelated work, exposes missing coverage, withdraws challenged explanations, and supports decisions without requiring surveillance or excessive coordination overhead.

8. Trust, correction, and the cost of another layer

The same connections that make a shared account useful can amplify error. A mistaken assumption reused across several teams is worse than an isolated incorrect summary. The proposed design therefore needs explicit treatment of correction, access, source quality, and interruption costs.

Correction must have consequences

Correcting wording in one panel is not enough. The system should identify affected interpretations, advice, saved views, and queued work; invalidate or qualify what no longer holds; and preserve the reason for the change. It should distinguish a personal perspective from a change to a shared record. Where competing accounts remain, it should not choose silently merely because one speaker has a more senior title.

This requires a traceable dependency model and a policy for derived information. It also requires humility about coverage: there may be work or evidence the system cannot see. Displaying a confidence score does not compensate for an unknown source boundary. Confidence in an interpretation and coverage of the relevant environment are different questions.

Access must follow the content and its consequences

Connecting an application does not imply permission to expose all its contents to all colleagues. Source-level access, shared-account permissions, retention, and derivative disclosure need explicit handling. Revocation must affect cached excerpts and generated accounts, not only future retrieval. Aggregation can reveal sensitive patterns even when individual snippets appear harmless; higher-level views need their own scrutiny.

On departure, required access removal must not wait for a complete handoff. An organization's retained work context and a person's portable learning are separate design questions. Personal portability is a longer-term possibility, not permission to export employer material, private notes, or restricted reasoning records. Ownership and applicable policy would need to govern any such transfer.

Source content is not an instruction channel

Messages and documents may contain mistaken, hostile, or manipulative content. Retrieval cannot grant tools new privileges. An AI participant must remain bound to its task, allowed actions, and review requirements. A source claiming that approval has been granted must be checked against the applicable authority basis; it cannot approve itself.

Ordinary work must stay ordinary

Governance should be proportional. Exploration, note drafting, and discussing alternatives should not become approval ceremonies. Consequential actions need stronger controls, but a person should see the specific missing condition and the shortest legitimate route to resolve it. The system should avoid repeated questions when a valid answer is already available.

Phossil must also justify its attention demands. An alert may be accurate yet not worth interrupting someone. Unhelpful suggestions, correction work, duplicated maintenance, and time spent inspecting the system belong in the cost of use. The goal is a more capable way of working, not a maximally populated graph.

9. How to test the thesis

The cases above define mechanisms that can be tested before broad effectiveness claims. Evaluation should move from controlled episodes to consented field use, retaining failures and costs at every stage. The following is a proposed protocol, not a completed study or committed numerical performance target.

Compare against a credible alternative

Use the same permitted source corpus, comparable task scopes, and the same underlying AI model where possible. The baseline should include the team's existing messaging, document, issue, and search tools configured competently, with normal handoff and review practices. Where feasible, include a capable existing persistent-memory or enterprise-context product with its normal features enabled. Do not strip competing products of their memory to create an artificial advantage. A separate assistant-only condition can help distinguish model assistance from persistent context, but cannot substitute for the strongest practical alternative.

For bounded tasks, randomize condition or counterbalance matched cases. Do not give the same person a case twice and treat their prior learning as product benefit. In field work, choose assignment at a team or episode level that limits spillover, record differences in experience and task difficulty, and report uncertainty rather than just averages. Shared learning is part of the product hypothesis but can also contaminate the comparison.

Measure the complete work, not the visible interface

- Decision and output quality: use a predeclared rubric and independent reviewers, blinded to condition where practical. Include whether the output respects constraints and acknowledges unresolved questions.
- Total effort: record preparation, search, interaction, source verification, correction, review, and rework. Distinguish active human time, elapsed delivery time, and concurrent AI activity.
- Evidence traceability: among a sampled set of material claims, count those whose cited accessible source actually supports the stated scope and time. A valid URL alone does not count as support.
- Correction propagation: introduce known corrections and inspect every identified affected output after revision and resume. Report corrected outputs over affected outputs, together with missed dependencies and newly introduced errors.
- Responsible action: record attempted and completed unauthorized actions separately. Missing authority, revoked access, and hostile source content should be deliberately tested; any successful unauthorized action is a critical failure in the test suite.
- Outcome follow-through: among decisions with an observable, due outcome, count those with attributable follow-up. Keep decisions without an observable outcome and those not yet due in separate categories.
- Reuse quality: test both a similar later episode and a superficially similar episode where the earlier reasoning should not apply. Reward appropriate non-reuse as well as useful transfer.
- Attention and operating cost: count dismissed or disruptive suggestions, maintenance effort, model and infrastructure expense, and human review cost. Report these alongside benefits, not in a separate footnote.

Sample size should be chosen after specifying the primary outcome, acceptable uncertainty, expected variance, and assignment unit. This paper does not invent a universal minimum sample. Exploratory pilot results should remain exploratory; confirmatory claims require a design that can support them.

Predeclare what would change our mind

Before a pilot, choose the quality floor, safety constraints, maximum acceptable overhead, and primary benefit sought with the participating team. Phossil should not count as successful merely because it improves one convenient metric after seeing the results. If quality falls, users spend more total effort, or gains disappear against a competent baseline, the product mechanism needs revision.

Controlled tests should include stale sources, contradictory accounts, unavailable evidence, partial tool failure, permission revocation, and a user changing their mind. An untouched comparison path matters as much as a path corrected during development. Reopening an episode should preserve exact constraints, not just a plausible summary.

Test business value separately from technical validity

Accurate reconstruction is necessary but does not establish demand. Research must examine who benefits, who maintains the account, who can sponsor access, who pays, and whether value persists after novelty wears off. Measure actual willingness to continue under disclosed operating costs and responsibilities, not just positive reactions to a demonstration.

The broad platform ambition should be tested through reuse: does an understanding developed in one episode improve a genuinely different later activity? A product that helps one workflow may still be valuable, but that result alone does not validate an organization-wide operating and learning layer.

10. Implications and boundaries

Phossil is proposed as a capability people and organizations can continually build on: not only a place to find information, but a way to inspect understanding, use it in work, and improve it through experience. Its breadth should come from a coherent mechanism that serves different activities, not from promising a separate untested feature for every role.

The three cases illustrate that mechanism across opportunity development, AI-assisted execution, and ambiguous organizational coordination. They intentionally preserve existing tools and human judgment. None requires that all work become a formal decision process or that every user enter a complex canvas. The depth of interaction should follow the work.

The unresolved engineering and product questions are substantial: reliable source interpretation, low-friction correction, propagation across derived outputs, scope-sensitive authority, access revocation, credible outcome observation, and justified reuse. The unresolved commercial questions are equally important: adoption, maintenance burden, buyer, persistence of value, and advantage over rapidly developing alternatives.

A defensible claim today is therefore a thesis and a test agenda, not a productivity guarantee. Organizations create understanding through work. Phossil aims to make that understanding something people, teams, and AI can keep using, questioning, and improving - so experience can make future work more capable. The next proof must come from observed use against credible alternatives.

Research method and disclosure

This is a selective evidence review, not a systematic literature review, meta-analysis, peer-reviewed publication, or YC endorsement. Sources were selected for relevance to assistance, collaboration, organizational learning, risk management, and overlapping software approaches. Primary research and first-party product documentation were prioritized. Product pages establish vendor-described capabilities, not independent performance. Material web sources were inspected in September 2026; study versions and access limitations are identified below.

The cases are authored synthetic scenarios. They contain no customer outcome data and do not represent deployments at Visa or another named company. No Phossil efficacy estimate, market-size calculation, or customer-validated return on investment is reported. Measures and safeguards are proposed requirements. Their appearance here does not establish implementation, security verification, or production availability.

Phossil is the proponent and publisher of this paper and has an interest in the proposed approach. Research and drafting were AI-assisted and require founder review before external distribution. HBR reading linked from the website is separate external commentary, not the source or publisher of this paper. Remaining research should include customer observation, additional organizational-learning literature, and independent evaluation of comparable products.

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